

$$\begin{array}{ccc}
 \mathbb{C} \setminus K & \xrightarrow{f} & \mathbb{C} \setminus K \\
 \downarrow \phi & & \uparrow \phi^{-1} \\
 \mathbb{C} \setminus \overline{\mathbb{D}} & \xrightarrow{z \mapsto z^n} & \mathbb{C} \setminus \overline{\mathbb{D}} \\
 \uparrow \psi & & \uparrow \psi \\
 \mathbb{H}^+ & \xrightarrow{w \mapsto nw} & \mathbb{H}^+
 \end{array}$$